# AI-Orchestrating Platform for Drug Repurposing


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## Motivation

Our goal is to accelerate and streamline drug repurposing through an AI-orchestrated, multi-model platform.

- **Drug development is slow and costly**, with high attrition rates in clinical trials.
- **Drug repurposing offers a faster route**, but current approaches are fragmented and inefficient.
- **Existing AI tools are siloed**, lacking orchestration across diverse data and methods.
- **A unified, multimodal platform is needed** to integrate structure, gene expression, knowledge graphs, and clinical evidence.
- **Bridging dry and wet labs** is essential to close the gap between computational discovery and experimental validation.

## Challenges

- **Integration complexity:** unifying diverse data types (proteins, small molecules, genes, literature) and modalities (text, graphs, sequences, 3D structures).
- **Scalability issues**: orchestrating multiple AI models efficiently across large and heterogeneous datasets.
- **Workflow orchestration**: aligning structure-based, knowledge-driven, and gene-expression pipelines into one seamless system.
- **Interoperability**: ensuring compatibility across AI techniques (ML, DL, NLP, LLMs).

## Key Highlights

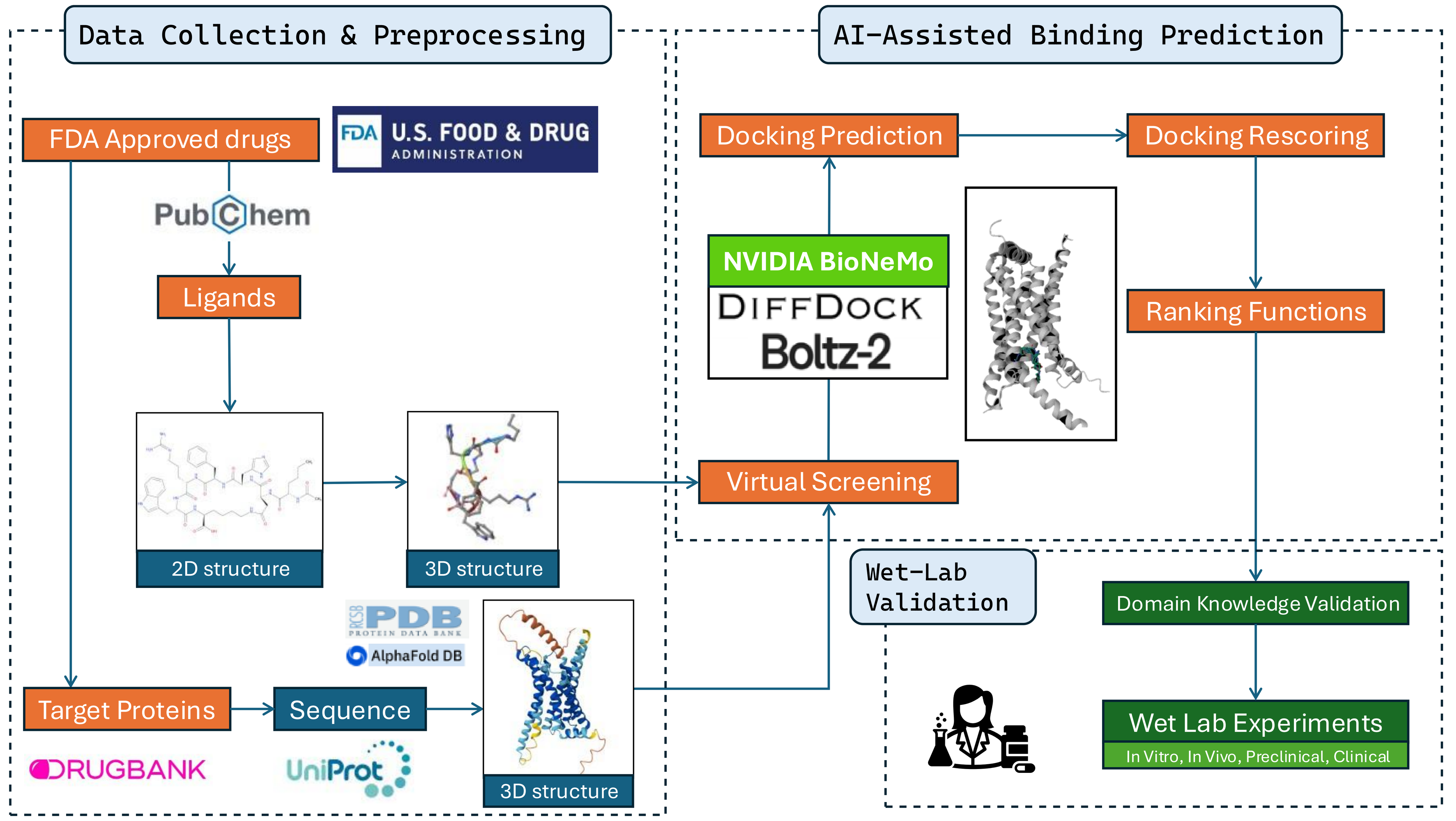
### Platform Strengths

- **First AI-orchestrated platform** for drug repurposing, integrating multimodal data and methods.
- **Powered by NVIDIA BioNeMo and NVIDIA cuGraph,** accelerating multi-model AI pipelines to rapidly screen thousands of drug–target pairs within hours.
- **Case study: Bremelanotide**, demonstrating successful discovery of a new therapeutic indication (Patent pending).

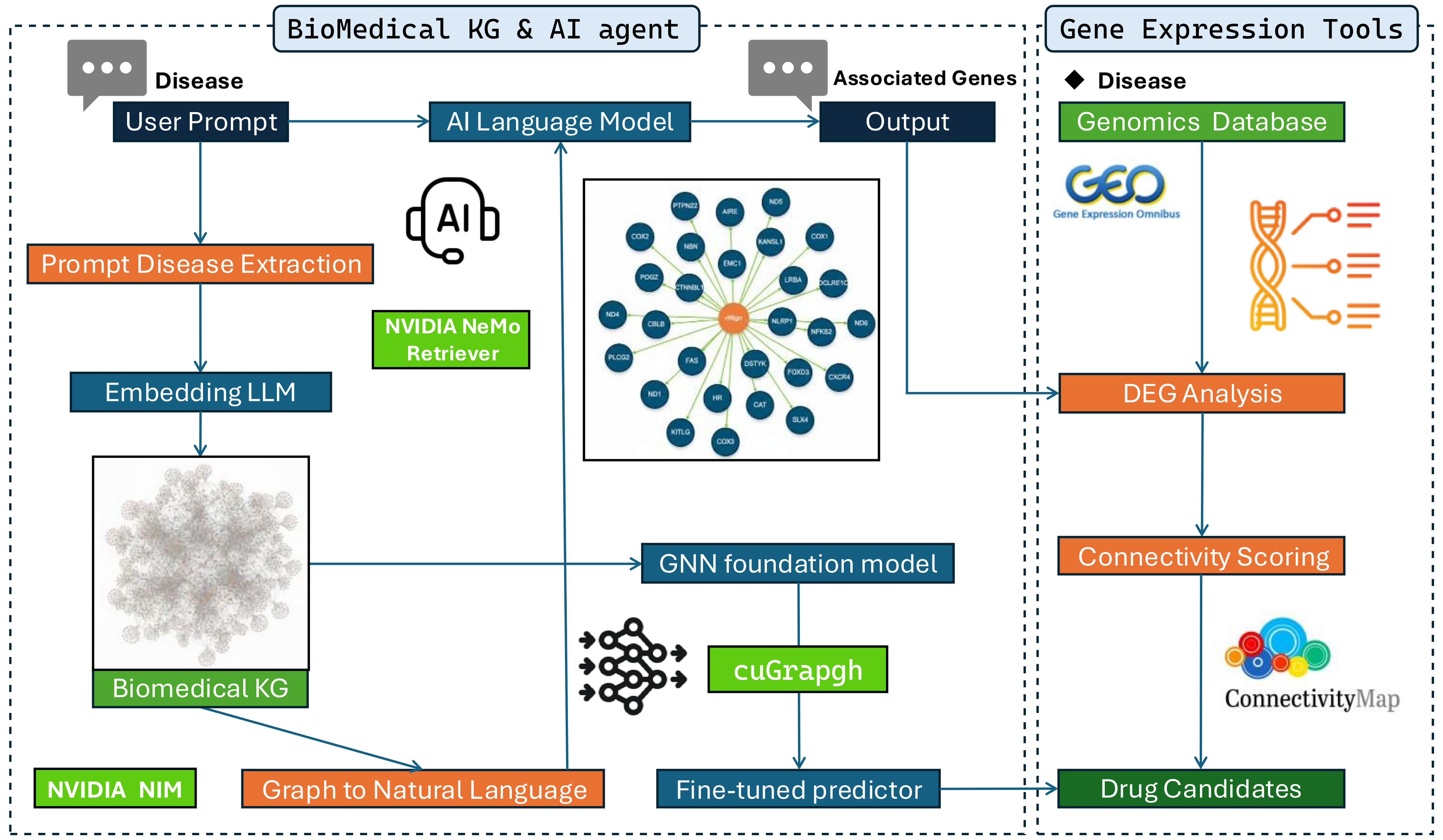
### Dual Workflows

- **Drug-Target Driven**: From **Drug → Disease** through structure-based screening and AI-powered binding prediction for small molecules and proteins.
- **Disease–Gene Driven**: From **Disease → Drug** via AI agents operating on knowledge graphs and retrieval-augmented generation (RAG), supported by gene expression analysis.

## Drug-Target Driven (Drug → Disease)




## Disease-Gene Driven (Disease → Drug)




## AI-Assisted Binding Prediction

**The tables demonstrate how NVIDIA BioNeMo aids in repurposing Bremelanotide:**

**DiffDock** for target docking and pose prediction
**Boltz2** for binding likelihood and affinity prediction

**Boltz2**

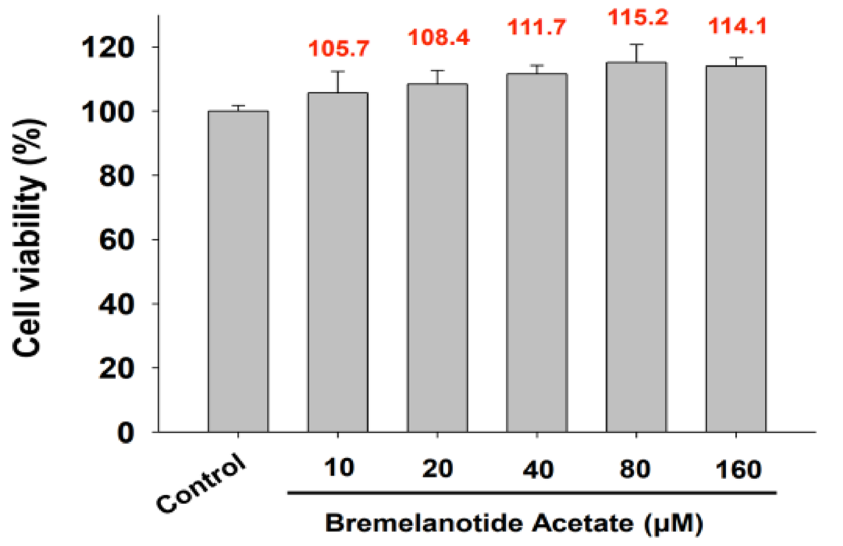
| Drug | Target | confidence | pIC50 | logIC50 | Binding Prob. |
|---|---|---|---|---|---|
| BREMELANOTIDE | GPCR_1 | **0.845939** | **11.404851** | **-2.361328** | **0.687734** |
| | 4PM7_1 | 0.844177 | 9.019709 | -0.612690 | 0.615548 |
| | 3VHU_1 | 0.708079 | 8.671108 | -0.357117 | 0.511958 |
| | 4NTX_1 | 0.656728 | 8.888714 | -0.516653 | 0.632282 |
| | ⋮ | ⋮ | ⋮ | ⋮ | ⋮ |
| HISTAMINE | GPCR_1 | 0.820489 | 6.475383 | 1.252651 | 0.280430 |
| | 4PM7_1 | 0.941027 | 5.515185 | 1.956609 | 0.377265 |
| | 3VHU_1 | 0.815950 | 5.861539 | 1.702684 | 0.287462 |
| | 4NTX_1 | 0.641424 | 6.737264 | 1.060657 | 0.343843 |
| | ⋮ | ⋮ | ⋮ | ⋮ | ⋮ |
| ⋮ | ⋮ | ⋮ | ⋮ | ⋮ | ⋮ |

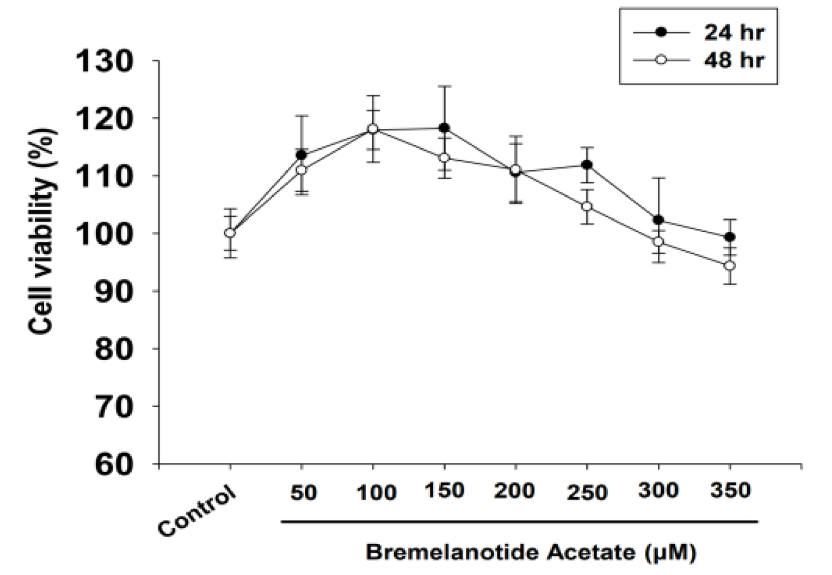
**DiffDock**

| Target | Drug | Pose Rank | Confidence | Best Pose |
|---|---|---|---|---|
| GPCR_1 | BREMELANOTIDE | Pose_rank01 | -1.649114 | |
| | | Pose_rank02 | -1.971637 | |
| | | Pose_rank03 | -2.086164 | |
| | | ⋮ | ⋮ | |
| | | Pose_rank10 | -3.214103 | |
| | HISTAMINE | Pose_rank01 | 0.370085 | |
| | | Pose_rank02 | 0.349616 | |
| | | Pose_rank03 | 0.322107 | |
| | | ⋮ | ⋮ | |
| | DOXEPIN | Pose_rank01 | -0.407854 | |
| | | Pose_rank02 | -0.538920 | |
| | | ⋮ | ⋮ | |
| ⋮ | ⋮ | ⋮ | ⋮ | ⋮ |

## Results: case study – Bremelanotide

**Bremelanotide Cytotoxicity in Pigment-Producing Cell Model: Repurposing Potential for Autoimmune Skin Condition**




- Bremelanotide Acetate (10–160 μM): **Cell viability increased to 105 –115%.**
- Higher concentrations (up to 350 μM): **No significant toxicity observed.**
- 24h and 48h treatments both showed **maintained viability** and **no cell death trend** even up 350 μM.

## References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